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PSScriptAnalyzer passing | Create Release passing | Check kubernetes App failing

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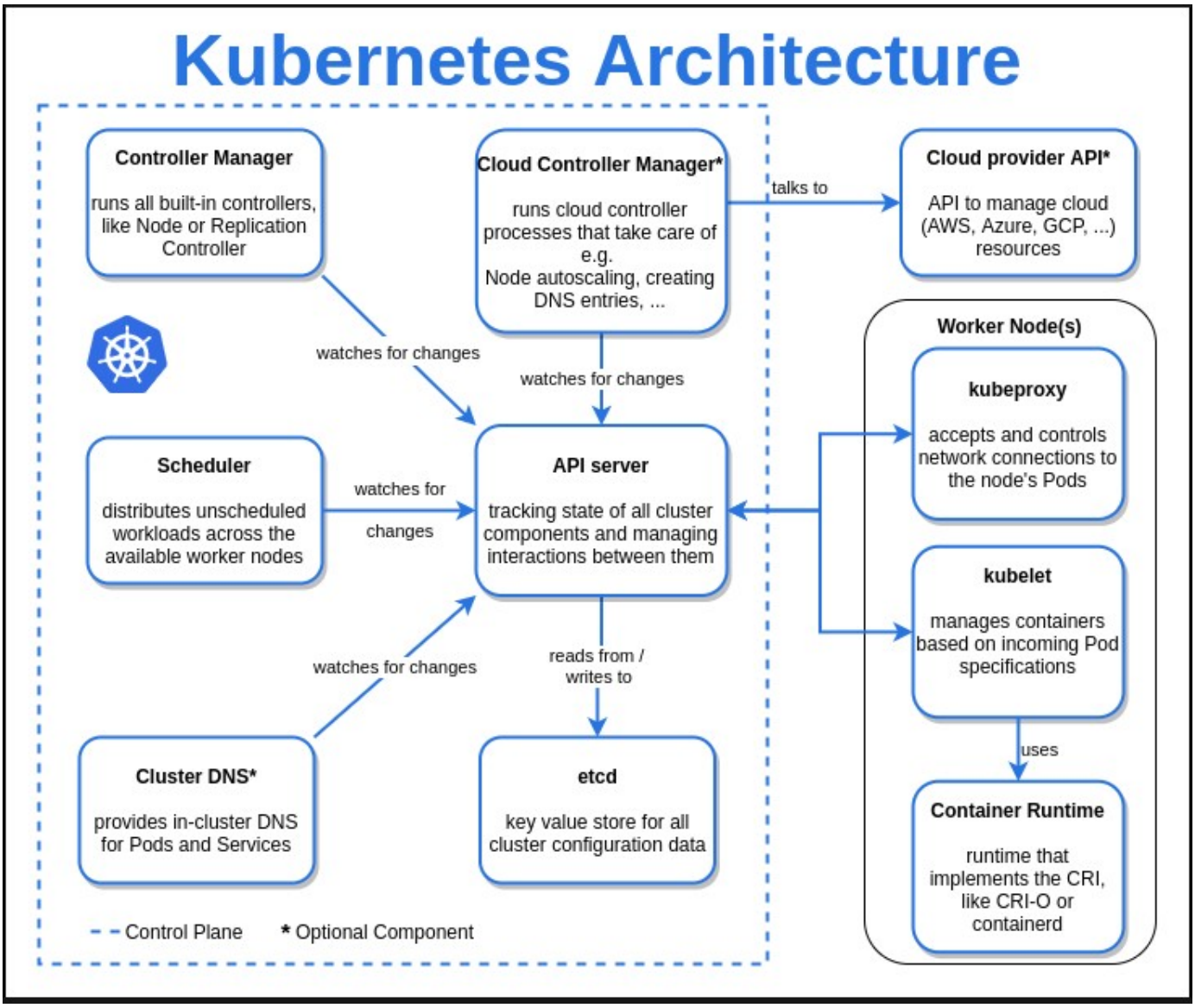
LINKEDIN

LEARNING KUBERNETES



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Summary

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About Project

This project aims to help students or professionals to learn the main concepts of kubernetes

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Getting Started

This is an example of how you may give instructions on setting up your project locally. To get a local copy up and running follow these simple example steps.

Prerequisites

This is an example of how to list things you need to use the software and how to install them.

- git
- Virtual Box and extension
- Vagrant

Installation

Clone repository

```
git clone https://github.com/marcossilvestrini/learning-kubernetes.git
```

Set ssh keys in security folder

```
# generate ssh key pair for your user access hosts
ssh-keygen -q -t ecdsa -b 521 -N '' -f ~/.ssh/id_ecdsa <<<y >/dev/null 2>&1
cp ~/.ssh/id_ecdsa.pub security/

# generate ssh key pair for rancher
ssh-keygen -q -t ecdsa -b 521 -N '' -f security/rancher-key-ecdsa <<<y >/dev/null 2>
cat security/rancher-key-ecdsa.pub >security/authorized_keys
```

Set network

Set network configuration for each VM in Vagrantfile. Example:

```
# NETWORK
    ol9_server01.vm.network "public_network", nic_type: "virtio", mac: "080027f3066a
      "Intel(R) I211 Gigabit Network Connection",
      "MediaTek Wi-Fi 6 MT7921 Wireless LAN"
    ]
```

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Set VM's resources

Set resource configuration as CPU, Memory,etc for each VM in Vagrantfile. Example:

```
# PROVIDER
infra_server01.vm.provider "virtualbox" do |vb|
    vb.linked_clone = true
    vb.name =VM_INFRA_SERVER01
    vb.memory = 2048
    vb.cpus = 1
end
```

Up kubernetes cluster

```
cd learning-kubernetes/vagrant/linux
vagrant up
```

Usage

Use this repository for get learning about kubernetes exam

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Roadmap

- [x] Create repository
- [x] Create github action for automation tasks
- [ ] Install kubernetes cluster

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kubernetes Engine work with namespaces(PID,NET,IPC,MNT,UTS) and cgroups.

Kubernetes ports

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CONTROL PLANE

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	6443*	Kubernetes API server	All
TCP	Inbound	2379-2380	etcd server client API	kube-apiserver, etcd
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10251	kube-scheduler	Self
TCP	Inbound	10252	kube-controller-manager	Self

WORKERS

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	30000-32767	NodePort	Services All

Font: https://livro.descomplicandokubernetes.com.br/pt/day_one/

Install kubernetes

Minikube

```
# install
curl -Lo minikube https://storage.googleapis.com/minikube/releases/latest/minikube-l
chmod +x ./minikube
sudo mv ./minikube /usr/local/bin/minikube

# get version
minikube version

# set hypervisor
minikube config set driver <YOUR_HYPERVISOR>

# up without hypervisor
minikube start --driver=hyperkit

# create cluster
minikube start --nodes 3 -p multinode-cluster

# get status of cluster
minikube status

# get ip address
```

```
minikube ip

# access minikube host
minikube ssh

# dashboard
minikube dashboard

# logs
minikube logs

# delete cluster
minikube delete
minikube delete --purge
```

Kind

```
# Install
curl -Lo ./kind https://kind.sigs.k8s.io/dl/v0.14.0/kind-linux-amd64
chmod +x ./kind
sudo mv ./kind /usr/local/bin/kind

# create cluster
kind create cluster
kind create cluster --name silvestrini

# get clusters
kind get clusters

# delete clusters
kind delete clusters $(kind get clusters)

## create yaml
cat << EOF > $HOME/kind-3nodes.yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
nodes:
  - role: control-plane
  - role: worker
  - role: worker
EOF

# create cluster
kind create cluster --name kind-multinodes --config $HOME/kind-3nodes.yaml
```

Reference: https://livro.descomplicandokubernetes.com.br/pt/day_one/

Kubectl

Install

```
# install
curl -LO https://storage.googleapis.com/kubernetes-release/release/`curl -s \
https://storage.googleapis.com/kubernetes-release/release/stable.txt`/bin/linux/amd6
chmod +x ./kubectl
mv ./kubectl /usr/local/bin/kubectl

# get version
kubectl version  --output=yaml --client

# kubectl autocomplete
source <(kubectl completion bash)

# kubectl alias
alias k=kubectl
complete -F __start_kubectl k
```

Commands

```
# list all resources
kubectl get all

# get namespaces
kubectl get namespaces

# describe namespaces
kubectl describe namespaces

# list nodes
kubectl get nodes
kubectl get nodes -o wide

# delete node
kubectl drain <node_name> --ignore-daemonsets --delete-emptydir-data
kubectl delete node <node_name>

# list pods
kubectl get pods

# list all pods
kubectl get pods --all-namespaces
kubectl get pods -A
kubectl get pods -A -o wide

# list pods in  kube-system namespace
kubectl get pod -n kube-system

# list pods with specif output
kubectl get pods -n kube-system -o yaml
kubectl get pods -n kube-system -o json

# describe details of pods
kubectl describe pod nginx
```

```
# create pod with manifest
kubectl apply -f pod-template.yaml
kubectl create -f pod.yaml

# execute pods
kubectl run nginx --image nginx

# create manifest|template
kubectl run my-nginx  --image nginx --port 80 --dry-run=client -o yaml >pod-template

# delete pods
kubectl delete pod nginx
kubectl delete -f pod-template.yaml

# create Service | expose pod
kubectl expose pod my-nginx

# list services
kubectl get services
kubectl get svc -o wide

# list services in system namespace
kubectl get svc -n kube-system

# delete service
kubectl delete service nginx

# connect in container
kubectl attach silvestrini -c infra

# execute command in container
kubectl exec infra ls
kubectl exec silvestrini -c infra -- ls
kubectl exec silvestrini -c infra -it sh

# get logs
kubectl logs my-nginx
kubectl logs -f my-nginx
```

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Contributing

Contributions are what make the open source community such an amazing place to learn, inspire, and create. Any contributions you make are **greatly appreciated**.

If you have a suggestion that would make this better, please fork the repo and create a pull request. You can also simply open an issue with the tag "enhancement". Don't forget to give the

project a star! Thanks again!

1. Fork the Project
2. Create your Feature Branch (`git checkout -b feature/AmazingFeature`)
3. Commit your Changes (`git commit -m 'Add some AmazingFeature'`)
4. Push to the Branch (`git push origin feature/AmazingFeature`)
5. Open a Pull Request

License

- This project is licensed under the MIT License * see the LICENSE.md file for details

Contact

Marcos Silvestrini - marcos.silvestrini@gmail.com
Twitter

Project Link: https://github.com/marcossilvestrini/learning-kubernetes

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Acknowledgments

- CNCF - Cloud Native Computing Foundation
- OCI - Open Container Initiative
- Borg
- kubernetes Website
- Github
- Issues
- CKA Certification
- CKAD Certification
- CKS Certification
- Book Linuxips
- Kind
- Minikube
- k0s
- k3s
- RKE2
- Up RKE2 Cluster HA

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